Instrumental Variables

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Introduction

- This lecture covers **instrumental variables** (IV) methods
- IV has been at the center of applied microeconometric research for the last few decades
- IV methods pioneered by economists have been widely adopted in other areas of empirical research
- We will begin by studying IV methods in a constant effects framework, then circle back to reinterpret these methods in a world of heterogeneous effects

IV History

- Instrumental variables methods were developed to deal with two seemingly unrelated problems
- Philip and Sewall Wright, 1920s:
  - Sought to estimate supply and demand relationships with data on equilibrium prices/quantities
  - This line of inquiry developed into the study of simultaneous equations models (SEMs)
- Wald, 1940: Sought to estimate a regression line when the regressor is measured with error
- Today IV is most often used as a method to eliminate omitted variables bias in single-equation causal models

Outline

1. Part 1: Constant Treatment Effect
   - Simple IV: 1 instrument, 1 endogenous variable
   - Multiple Instruments: 2SLS
   - IV as GMM
   - Overidentification Tests
   - 2SLS Mistakes
   - Pop-Quiz!
   - Weak IV (Appendix)
2. Part 2: Heterogenous Treatment Effects
   - LATE Theorem
   - Compliers Characteristics
   - Over-id tests with HTE
3. Part 3: IV and Sample Selection Models
   - Heckit
   - Control Function
   - Lee Bounds

Main reference: Mostly Harmless Econometrics.

Section 1

Constant Effects

Outline

1. Constant Effects
   - Simple IV
   - 2SLS and Multiple Instruments
   - IV as GMM
   - Overidentification Tests
   - Imperfect Compliance
   - 2SLS Mistakes

A Simple Causal Model

- Consider a simple constant-effects causal model:

$$Y_i(s) = \alpha + \rho s + \epsilon_i$$

- $Y_i(s)$ is i's potential wage if s/he attends school for s years
- This additively separable function form implies constant causal effects: $Y_i(s) - Y_i(s-1) = \rho \ \forall (i, s)$
- We observe schooling S_i and earnings $Y_i(S_i)$

A Simple Causal Model

- Equation for observed earnings:

$$Y_i = \alpha + \rho S_i + \epsilon_i$$

- This is a structural equation, not a regression, so no guarantee that $Cov(S_i, \epsilon_i) = 0$
- Project ϵ_i on to unobserved ability A_i:

$$\epsilon_i = \gamma A_i + v_i$$

- $Cov(A_i, v_i) = 0$ by definition; assume $Cov(S_i, v_i) = 0$
- Unobserved ability A_i is a potential source of omitted variable bias

Long and Short Regressions

- "Long" regression:

$$Y_i = \alpha + \rho S_i + \gamma A_i + v_i$$

- "Short" regression:

$$Y_i = \alpha_s + \rho_s S_i + v_{is}$$

- OVB formula:

$$\rho_s = \rho + \gamma \frac{Cov(S_i, A_i)}{Var(S_i)}$$

- The regression we've got (short) is not the regression we want (long)
- IV is a method for obtaining the long regression coefficient without data on the omitted regressor A_i

Instrumental Variables

$$Y_i = \alpha + \rho S_i + \epsilon_i$$

- Suppose we find an **instrument**, Z_i, that satisfies two conditions:
  1. **First stage:** $Cov(S_i, Z_i) \neq 0$
  2. **Exclusion restriction:** $Cov(\epsilon_i, Z_i) = 0$
- The first stage condition requires the instrument to be correlated with the causal variable of interest. This is testable.

Instrumental Variables

$$Y_i = \alpha + \rho S_i + \epsilon_i$$

1. **First stage:** $Cov(S_i, Z_i) \neq 0$
2. **Exclusion restriction:** $Cov(\epsilon_i, Z_i) = 0$

- The exclusion restriction requires the instrument to be uncorrelated with the structural error ϵ_i. This requires
  - Z_i must be uncorrelated with all unobserved determinants of earnings ("as good as randomly assigned")
  - Z_i has no direct causal impact on Y_i
- The exclusion restriction is not testable (caveats later)

Instrumental Variables

- The exclusion restriction implies

$$Cov\left(Y_i, Z_i\right) = Cov\left(\alpha + \rho S_i + \epsilon_i, Z_i\right)$$

$$= \rho Cov(S_i, Z_i)$$

- If the first stage assumption is satisfied, we can divide by $Cov(S_i, Z_i)$ to obtain

$$\frac{Cov\left(Y_i, Z_i\right)}{Cov\left(S_i, Z_i\right)} = \rho$$

- The ratio of the covariance of the outcome with the instrument to the covariance of the endogenous variable with the instrument identifies the long regression coefficient ρ
- This ratio of covariances is the population **instrumental variables** coefficient

Instrumental Variables

$$\frac{Cov\left(Y_i, Z_i\right)}{Cov\left(S_i, Z_i\right)} = \rho$$

- Divide numerator and denominator by $Var(Z_i)$ to obtain

$$\frac{Cov(Y_i, Z_i) / Var(Z_i)}{Cov(S_i, Z_i) / Var(Z_i)} = \rho$$

- What is the interpretation of this expression?

Instrumental Variables

$$\frac{Cov(Y_i, Z_i)/Var(Z_i)}{Cov(S_i, Z_i)/Var(Z_i)} = \rho$$

- The IV coefficient can be viewed as a ratio of two OLS regression coefficients:

$$Y_i = \lambda + \delta Z_i + u_i$$

$$S_i = \kappa + \pi Z_i + \eta_i$$

- δ is the **reduced form** coefficient, and π is the **first stage** coefficient
- The IV coefficient is $\rho_{IV} = \delta/\pi$, the reduced form divided by the first stage
- If our two IV assumptions are satisfied, then $\rho_{IV} = \rho$: IV recovers the long regression coefficient

IV Example: Angrist and Krueger (1991)

- IV eliminates omitted variable bias and allows us to recover the causal parameter of interest
- But where did the instrument Z_i come from?
- Good instruments typically come from institutional knowledge combined with plausible assumptions about behavioral relationships
- Well-known example: Angrist and Krueger (1991) study of the returns to education

Angrist and Krueger (1991)

- AK estimate the causal effect of schooling on earnings
- AK's IV strategy is motivated by the interaction between compulsory schooling and age-at-entry laws
  - Students can typically drop out of school on the day they turn 16
  - Birth date cutoff for starting age: Kids usually start school in the fall of the calendar year in which they turn six
- These rules generate differences in mean educational attainment by date of birth

Birth date	School start date	Dropout date	Schooling at dropout date
January 2, 1930	September 1, 1936	January 2, 1946	9.5 years
December 31, 1930	September 1, 1936	December 31, 1946	10.5 years

Angrist and Krueger (1991)

- The combination of dropout and age-at-entry rules means that compulsory schooling requirements are higher for those born late in the year
- This suggests an IV strategy using date of birth as an instrument for schooling
- AK use a "quarter of birth" instrument, an indicator for being born early in the year (January - March)
- What are the first stage and exclusion restriction assumptions in this example?

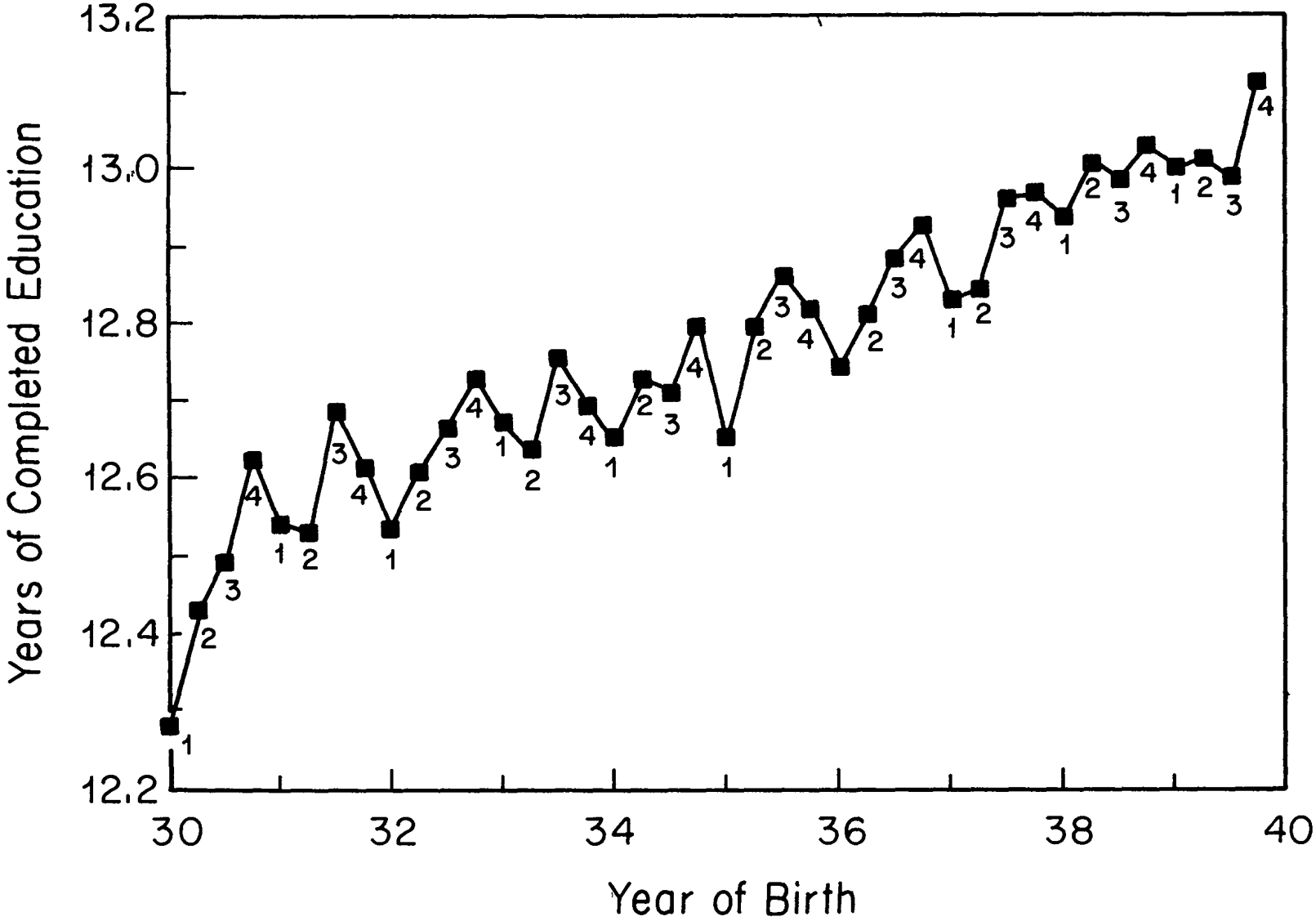


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

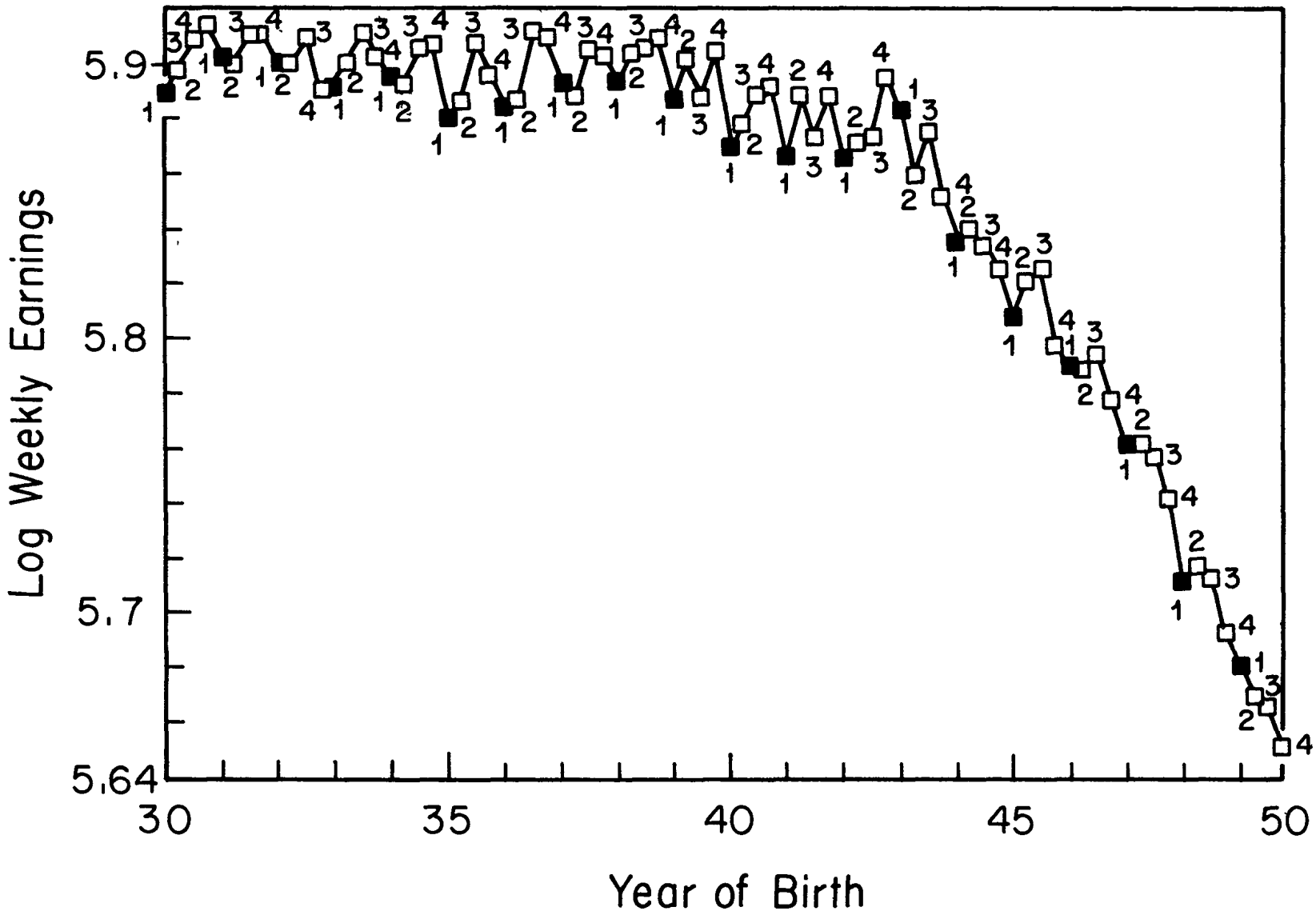


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

The Wald Estimator

$$\rho_{IV} = \frac{Cov(Y_i, Z_i)/Var(Z_i)}{Cov(S_i, Z_i)/Var(Z_i)}$$

- When Z_i is binary, the first stage and reduced form coefficients become differences in means:

$$\rho_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[S_i|Z_i = 1] - E[S_i|Z_i = 0]}$$

- This is the population analogue of the Wald (1940) estimator, the ratio of the mean difference in the outcome to the mean difference in the endogenous regressor

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) − (2)
ln (wkly. wage)	5.8916	5.9027	−0.01110
			(0.00274)
Education	12.6881	12.7969	−0.1088
			(0.0132)
Wald est. of return to education			0.1020
			(0.0239)
OLS return to education			0.0709
			(0.0003)

IV with Covariates

- When doing IV it's often necessary to control for additional observed variables
- Example: QOB is correlated with age, which likely also affects earnings. This suggests a violation of the exclusion restriction; we may want to control for smooth functions of age
- The long regression of interest now includes the control vector X_i:

$$Y_i = \alpha + \rho S_i + X_i'\gamma + \gamma A_i + v_i$$

- Control variables that are neither instruments nor instrumented are "exogenous covariates"

IV with Covariates

- Reduced form and first stage regressions augmented with covariates:

$$Y_i = \lambda + \delta Z_i + X_i' \psi + u_i$$

$$S_i = \kappa + \pi Z_i + X_i' \phi + \eta_i$$

- As before we can solve for ρ as δ / π
- Applying the Frisch-Waugh-Lovell theorem, this can be written

$$\frac{\delta}{\pi} = \frac{Cov\left(Y_i, \tilde{Z}_i\right)}{Cov\left(S_i, \tilde{Z}_i\right)}$$

Indirect Least Squares

$$\frac{\delta}{\pi} = \frac{Cov\left(Y_i, \tilde{Z}_i\right)}{Cov\left(S_i, \tilde{Z}_i\right)}$$

- The sample analogue of this ratio is called the Indirect Least Squares (ILS) estimator
- Controlling for covariates is equivalent to using the residual from a regression of Z_i on X_i as the instrument
- The first stage and exclusion restriction assumptions now need to hold for $\tilde{Z}_i$ rather than Z_i
- Importantly, this means that the first stage relationship between S_i and Z_i must survive the addition of the controls X_i

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   - Simple IV
   - 2SLS and Multiple Instruments
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   - Imperfect Compliance
   - 2SLS Mistakes

Multiple Instruments

- Now suppose we have multiple instruments, $Z_{i1}...Z_{iJ}$, collected in the $J \times 1$ vector Z_i
- Example: dummies for each quarter of birth, rather than just the first quarter
- It's no longer clear how to solve for the long regression coefficient ρ. We could use any one of the J instruments, or any linear combination of them
- ρ is now **overidentified**: there are multiple ways to solve for the parameter of interest
- How should we combine the available information?

Multiple Instruments

- Our structural equation of interest is still

$$Y_i = \alpha + \rho S_i + X_i' \gamma + \epsilon_i$$

- Consider a first-stage regression of S_i on all the instruments and exogenous controls:

$$S_i = \kappa + Z_i' \pi + X_i' \phi + \eta_i$$

Multiple Instruments

- Substitute the first stage regression into the structural equation:

$$\begin{aligned} Y_i &= \alpha + \rho\left[\kappa + Z_i'\pi + X_i'\phi + \eta_i\right] + X_i'\gamma + \epsilon_i \\ &= \alpha + \rho\left(\kappa + Z_i'\pi + X_i'\phi\right) + X_i'\gamma + \left(\rho\eta_i + \epsilon_i\right) \\ &= \alpha + \rho\hat{S}_i + X_i'\gamma + \nu_i \end{aligned}$$

- Here $\hat{S}_i = \kappa + Z_i'\pi + X_i'\phi$ is the predicted value of S_i from the first stage
- The new residual is $\nu_i = \rho\eta_i + \epsilon_i = \rho(S_i - \hat{S}_i) + \epsilon_i$

Two-Stage Least Squares

$$Y_i = \alpha + \rho \hat{S}_i + X_i' \gamma + \nu_i$$

- $\hat{S}_i$ and X_i are uncorrelated with ν_i (why?), so OLS recovers the parameters of this equation
- This is the two-stage least squares (2SLS) estimator:
  1. Regress S_i on all the instruments and exogenous controls, and obtain predicted values $\hat{S}_i$
  2. Regress Y_i on $\hat{S}_i$ and the exogenous controls
- 2SLS provides one method for combining instruments. Do you think it is the most efficient?

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IV Estimation

- Suppose we have N observations on (Y_i, S_i, Z_i, X_i), where
  - S_i is a $K \times 1$ vector of endogenous variables (there now may be more than one)
  - Z_i is a $J \times 1$ vector of instruments
  - X_i is an $L \times 1$ vector of exogenous controls
- Let $\mathbf{X}_i = (1, X_i', S_i')'$ and $\mathbf{Z}_i = (1, X_i', Z_i')'$

IV as GMM

- Let's augment our structural equation

$$Y_i = \alpha + S_i'\rho + X_i'\gamma + \epsilon_i$$

  - S_i is a $K \times 1$ vector of endogenous variables (there now may be more than one)
  - Z_i is a $J \times 1$ vector of instruments ($J \geq K$)
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  - X_i is an $L \times 1$ vector of exogenous controls
  - Let $\mathbf{X}_i = (1, X_i', S_i')'$ and $\mathbf{Z}_i = (1, X_i', Z_i')'$

- We know (from either economic theory or institutional knowledge of Z_i and S_i) that

$$\mathrm{E}[\mathbf{Z}_i'\varepsilon_i] = \mathrm{E}[\mathbf{Z}_i'(y_i - \alpha - S_i'\rho - X_i'\gamma)] = \mathrm{E}[m_i(\theta)] = 0 \qquad (1)$$

  where $\theta = (\rho; \gamma)$

- What do we call this? How many equations? How many unknowns?

Casting everything as GMM

- Recall previous class on GMM (see Estimation slide deck). GMM minimizes the following criterion function

$$Q_{GMM}(\theta) = \mathrm{E}[m_i(\theta)' \mathbf{W} m_i(\theta)] \tag{2}$$

- What happens to $\mathbf{W}$ when $J = K$?

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- How do we choose $\mathbf{W}$ when $J > K$?
- Turns out that 2SLS minimizes (2) by setting $\mathbf{W} = E[\mathbf{Z}_i \mathbf{Z}_i']^{-1} \rightarrow$ When is 2SLS efficient? See Section 4.2.1 of MHE for full proof.

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- Turns out that 2SLS minimizes (2) by setting $\mathbf{W} = E[\mathbf{Z}_i \mathbf{Z}_i']^{-1} \rightarrow$ When is 2SLS efficient? See Section 4.2.1 of MHE for full proof.
- What is the optimal $\mathbf{W}$ in a general setup?

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Over-id Test

- GMM formulation also useful to think on how we can test the validity of the "model".
- When there are more instruments than endogenous variables ($J > K$) there are multiple ways to solve for the structural parameters.
- Essentially an over-id specification test (as introduced in the Estimation slide-deck) tells us how close to being satisfied the moment condition are.
- In the IV setup, this means that an overid test checks whether *all* the IV moment restrictions are close enough to being satisfied.

Overid Tests in the IV world

- Let $\hat{\theta}$ be the minimizer of the sample GMM criterion function

$$\hat{\theta} = \underset{c}{\operatorname{argmin}}\, \hat{Q}_{GMM}(c) = N\hat{m}_i(c)'\mathbf{W}\hat{m}_i(c) \tag{3}$$

  where $\hat{m}_i(\theta) = \frac{1}{N}\mathbf{Z}_i'(y_i - \alpha - S_i'\rho - X_i'\gamma)$ (evaluated at the "truth" θ)

- If $\mathbf{W} = \mathrm{E}[\sqrt{N}\hat{m}_i(\theta)\sqrt{N}\hat{m}_i(\theta)']^{-1}$, then as $N \to \infty$

$$\hat{Q}_{GMM}(\hat{\theta}) \sim \chi^2_{J-K} \tag{4}$$

  (sometimes this is also called the J-test in the IV setup).

- If $\hat{Q}_{GMM}(\hat{\theta})$ is large relative to critical value of $\chi^2_{J-K} \to$ model is rejected.

- Rejection means that one of the J instrument is false, i.e. there is one instrument for which $\mathrm{E}[Z_i\epsilon_i]$ is actually not zero!

Intuition

- Suppose that there are no controls and we have one endogenous variable S_i and two potential instruments Z_{i1} and Z_{i2}.
- Two instruments give us two potential (simple) IV estimates of ρ.
- First estimate of ρ obtained by solving moment condition

$$\mathrm{E}[Z_{i1}(y_i - S_i\rho_1)] = 0 \tag{5}$$

- And the second one is obtained by fitting

$$\mathrm{E}[Z_{i2}(y_i - S_i\rho_2)] = 0 \tag{6}$$

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- Over-id tests whether a model that imposes $\rho_2 = \rho_1$ can fit the data.
- What is the key underlying assumption that we need to make this test truly useful?

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Which research questions can be answered causally?

- Go back to our discussion we had in the first class: often times economists are interested in the causal effect of *choices* made by agents (e.g. attend a private university, smoking, do a pre-cancer screening test, etc).
- The gold standard in research is the RCT but sometimes you can't randomize individuals into treatment (IRB won't allow it and for very good reasons).
- Strategy is then to randomize Z_i where Z_i is some incentive to take up a given treatment of interest, D_i.
- Even if there is imperfect compliance $\Pr(D_i = 1|Z_i = 1) \neq 1$ one can use IV techniques to estimate a causal effect.
- Simple but powerful idea that still needs to make its way in *all* fields...

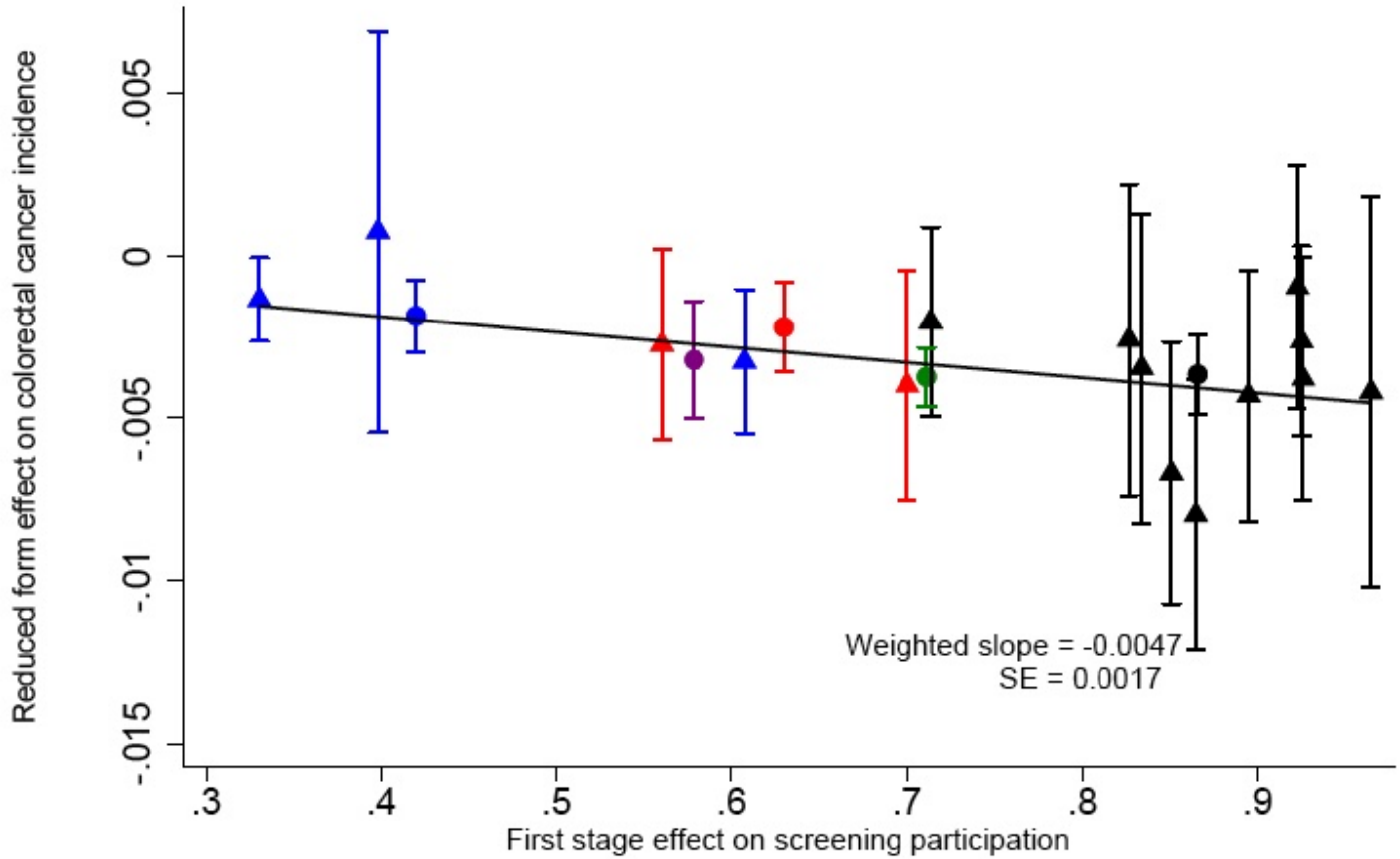
IV Makes Its Way in Clinical Trials

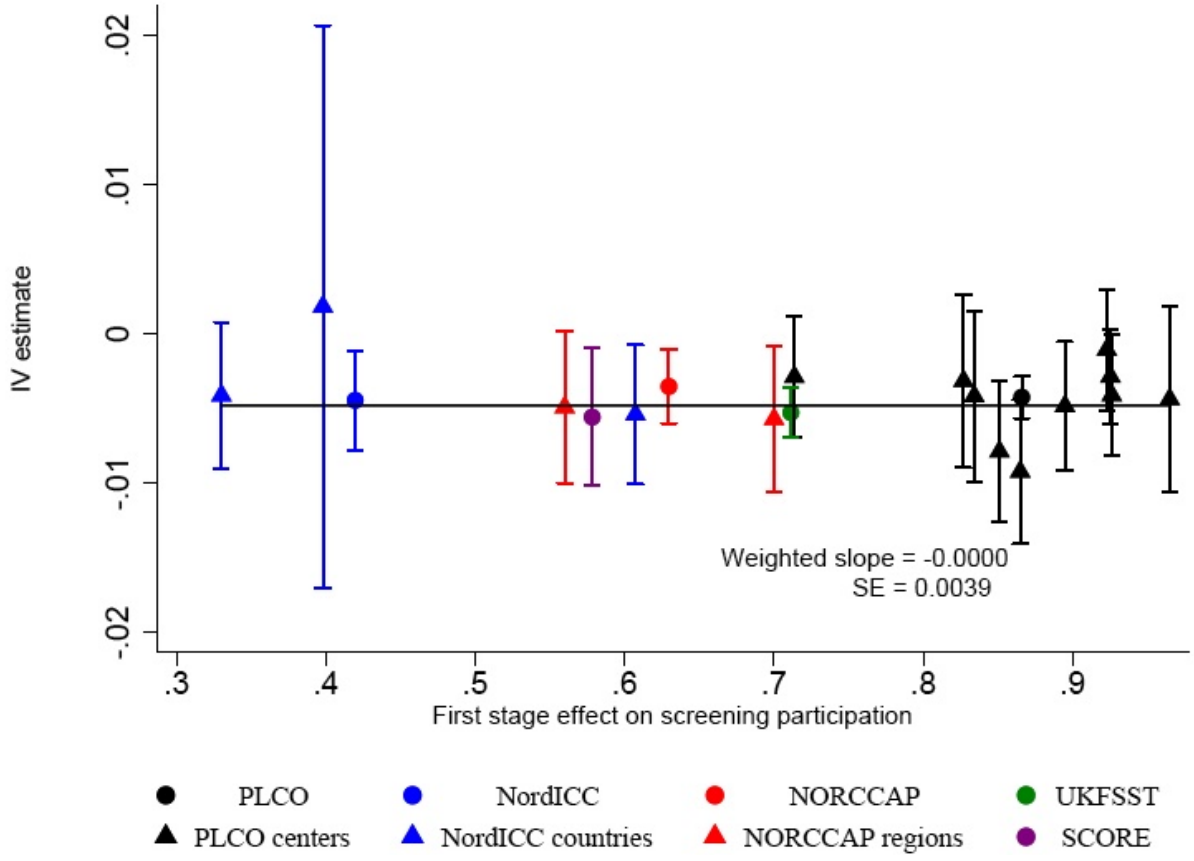
- RQ: do pre-cancer screenings improve health?
- Can't force doctors (or patients) to test so people have tried to answer this question by randomizing the opportunity to screen...
- Studies in this area thus report Intention-to-Treat (or Intention-to-Screen) Effects: $\mathrm{E}[Y_i|Z_i = 1] - \mathrm{E}[Y_i|Z_i = 0]$.
- There seems to be a lot of heterogeneity in the ITTs estimated across a whole variety of studies
- Winawer (2023) asks "are these intention-to-treat observations applicable to other clinical environments?"

IV Methods Reconcile Intention-to-Screen Effects Across Pragmatic Cancer Screening Trials
Joshua Angrist and Peter Hull
NBER Working Paper No. 31443
July 2023, Revised September 2023
JEL No. C21,C26,C93,I12

ABSTRACT

Pragmatic cancer screening trials mimic real-world scenarios in which patients and doctors are the ultimate arbiters of treatment. Intention-to-screen (ITS) analyses of such trials maintain randomization-based apples-to-apples comparisons, but differential adherence (the failure of subjects assigned to screening to actually get screened) makes ITS effects hard to compare across trials and sites. We show how instrumental variables (IV) methods address the nonadherence challenge in a comparison of estimates from 17 sites in five randomized trials measuring screening effects on colorectal cancer incidence. While adherence rates and ITS estimates vary widely across and within trials, IV estimates of per-protocol screening effects are remarkably consistent. An application of simple IV tools, including graphical analysis and formal statistical tests, shows how differential adherence explains variation in ITS impact. Screening compliers are also shown to have demographic characteristics broadly similar to those of the full trial study sample. These findings argue for the clinical relevance of IV estimates of cancer screening effects.


Reduced form effect on colorectal cancer incidence
.005
0
-.005
-.01
-.015
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0047
SE = 0.0017


IV estimate
.02
.01
0
-.01
-.02
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0000
SE = 0.0039
PLCO
NordICC
NORCCAP
UKFSST
PLCO centers
NordICC countries
NORCCAP regions
SCORE

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2SLS Mistakes

- 2SLS is the workhorse IV estimator used by empirical researchers
- It's worth covering some common pitfalls related to 2SLS estimation

2SLS Mistakes 1: Standard Errors

- It's usually not a good idea to run the two stages of 2SLS manually
- For one thing, this will give the wrong standard errors
- Consider a setting with one endogenous variable S_i

$$Y_i = \alpha + \rho S_i + \epsilon_i$$

- If you run 2SLS manually, the second stage is

$$Y_i = \alpha + \rho \hat{S}_i + v_i$$

- Here $v_i = \epsilon_i + \rho(S_i - \hat{S}_i)$
- Manual 2SLS will typically generate standard errors that are too large.
- The residual variance should be calculated from the structural residual $Y_i - \hat{\alpha} - \hat{\rho} S_i$, not the second-stage residual $Y_i - \hat{\alpha} - \hat{\rho} \hat{S}_i$

2SLS Mistakes 2: Covariate Ambivalence

- A second possible mistake in manual 2SLS is the treatment of covariates
- Suppose we run the 2SLS procedure

$$S_i = \kappa + Z_i'\pi + X_{1i}'\phi_1 + \eta_i$$

$$Y_i = \alpha + \rho\hat{S}_i + X_{1i}'\theta_1 + X_{2i}'\theta_2 + v_i$$

- The covariate vector X_{2i} is included in the second stage but not the first (the opposite of the instruments Z_i)
- What's wrong with this?

2SLS Mistakes 2: Covariate Ambivalence

$$S_i = \kappa + Z_i'\pi + X_{1i}'\phi_1 + \eta_i$$

$$Y_i = \alpha + \rho\hat{S}_i + X_{1i}'\theta_1 + X_{2i}'\theta_2 + v_i$$

- The second-stage residual is $v_i = \epsilon_i + \rho(S_i - \hat{S}_i)$
- Since $S_i - \hat{S}_i$ is a residual from a regression that includes X_{1i}, $Cov(X_{1i}, S_i - \hat{S}_i) = 0$
- But in general $Cov(S_i - \hat{S}_i, X_{2i}) \neq 0$, so we may have created a correlation between the error term and covariates, leading to bias in all parameter estimates
- "If a covariate is good enough for the second stage, it's good enough for the first."

2SLS Mistakes 3: The Forbidden Regression

- In response to repeated mistakes by graduate students, Jerry Hausman labeled one 2SLS mistake "the forbidden regression"
- The forbidden regression involves the erroneous application of 2SLS logic to nonlinear models
- Suppose there is a dummy endogenous variable $T_i \in \{0, 1\}$, a vector of covariates X_i and a vector of instruments Z_i
- The causal equation is

$$Y_i = \alpha + \rho T_i + X_i'\theta + \epsilon_i$$

2SLS Mistakes 3: The Forbidden Regression

- First stage regression:

$$T_i = \kappa + Z_i'\pi + X_i'\phi + \eta_i$$

- Since T_i is binary, $E[T_i|Z_i, X_i]$ is likely nonlinear unless the model is saturated. The first stage is therefore a misspecified model of the CEF
- One might be tempted to try a nonlinear model to better approximate the CEF, e.g. a probit:

$$Pr[T_i = 1|X_i, Z_i] = \Phi(\kappa + Z_i'\pi + X_i'\phi)$$

2SLS Mistakes 3: The Forbidden Regression

- The nonlinear first stage can be used to obtain predicted values:

$$\hat{T}_i = \Phi\left(\hat{\kappa} + Z_i'\hat{\pi} + X_i'\hat{\phi}\right)$$

- One might then be tempted to run a 2SLS-style second stage:

$$Y_i = \alpha + \rho\hat{T}_i + X_i'\theta + \epsilon_i$$

- This regression is forbidden!

2SLS Mistakes 3: The Forbidden Regression

- To see why the forbidden regression is forbidden, rewrite the structural equation

$$Y_i = \alpha + \rho \hat{T}_i + X_i'\theta + \left(\epsilon_i + \rho(T_i - \hat{T}_i)\right)$$

- If the first stage is estimated via OLS, then the residual $(T_i - \hat{T}_i)$ is guaranteed to be uncorrelated with the fit $\hat{T}_i$ and with the included covariates X_i
- Nonlinear residuals do not have this property. $Cov\left(T_i - \hat{T}_i, \hat{T}_i\right)$ and $Cov\left(T_i - \hat{T}_i, X_i\right)$ will generally only equal zero when the $E\left[T_i | X_i, Z_i\right]$ is really a probit
- Who knows whether the probit CEF is right? Better to stick with a linear first stage, which guarantees zero covariance between residuals and fits/regressors even if the CEF is misspecified

Pop-Quiz

Pop-Quiz

- Suppose you came up with what you think is a good instrument, Z_i.
- You are however not 100% sure about the exclusion restriction.
- Inspired by your knowledge of regression and the meaning of the exclusion restriction you come up with the following test for the validity of the instrument

$$y_i = \alpha + \beta S_i + \gamma Z_i + r_i \tag{7}$$

- That is you run a regression on your endogenous variable S_i and the instrument Z_i.
- Since the instrument only operates through the endogenous variable S_i, you argue that if the exclusion restriction is valid, then $\beta = \rho$ (the true casual effect) and $\gamma = 0$.
- Right? Wrong?

APPENDIX

IV in Finite Samples

- OLS is an unbiased estimator of population projection coefficients in finite samples (with a linear CEF or nonstochastic regressors)
- IV estimators do not have this property
- IV is a ratio of reduced form and first stage coefficients:

$$E\left[\hat{\rho}_{IV}\right] = E\left[\frac{\hat{\delta}}{\hat{\pi}}\right]$$

$$\neq \frac{E\left[\hat{\delta}\right]}{E\left[\hat{\pi}\right]}$$

$$= \rho$$

IV in Finite Samples: Example

- Consider the following reduced form and first stage:

$$Y_i = \lambda + \delta Z_i + u_i$$

$$S_i = \kappa + \pi Z_i + \eta_i$$

- Suppose the instruments are non-stochastic, and

$$(u_i, \eta_i) \sim N\left(0, \begin{bmatrix} \sigma_u^2 & \tau\sigma_u\sigma_\eta \\ \tau\sigma_u\sigma_\eta & \sigma_\eta^2 \end{bmatrix}\right)$$

- Then the finite-sample distribution of the reduced form and first stage coefficients is

$$(\hat{\delta}, \hat{\pi}) \sim N\left((\delta, \pi), \begin{bmatrix} \sigma_u^2 \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} & \tau\sigma_u\sigma_\eta \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} \\ \tau\sigma_u\sigma_\eta \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} & \sigma_\eta^2 \left[\sum_i (Z_i - \bar{Z})^2\right]^{-1} \end{bmatrix}\right)$$

IV in Finite Samples: Example

- The IV coefficient, $\hat{\rho} = \hat{\delta}/\hat{\pi}$, is a ratio of two jointly normal random variables
- $\hat{\rho}$ therefore follows a normal ratio distribution
- Normal ratio distributions do not have moments, so $E[\hat{\rho}]$, $Var(\hat{\rho})$, and all higher moments do not exist. In this sense $\hat{\rho}$ is biased
- If $\tau = 0$ and $\pi \to 0$, $\hat{\rho}$ becomes the ratio of two independent mean zero normals, which follows a Cauchy distribution (see Nelson and Starz 1990 for more)

IV in Finite Samples: Many/Weak Problems

- In general, just-identified IV estimators have no moments and fat tails
- However, just-id estimators are approximately centered in the right place in the sense that $Q_{0.5}(\hat{\rho}_{IV}) \approx \rho$ (more details later)
- It turns out that overidentified IV estimators are not centered at the right place: they are biased towards OLS
- The severity of finite-sample bias increases with the degree of overfitting AND as the first-stage relationship between instruments and endogenous regressors gets weaker
- IV bias is therefore worse when the instruments are "many/weak"
- Bound, Jaeger and Baker (1995) showed that this problem can be important in practice, spawning a large literature on dealing with many/weak instruments

Bound, Jaeger and Baker (1995)

- Bound, Jaeger and Baker (1995) revisit the Angrist/Krueger QOB instrument
- The Wald estimates we studied are just-identified, so the many/weak critique does not apply (a common misconception)
- However, AK also estimate overidentified 2SLS models of the form

$$Y_i = \rho S_i + \sum_{b=1}^{50} \theta_b B_{ib} + \sum_{t=1}^{30} \psi_t T_{it} + \epsilon_i$$

$$S_i = \sum_{q=1}^{3} \left[\sum_{b=1}^{50} \pi_{qb} B_{ib} Q_{iq} + \sum_{t=1}^{30} \tau_{qt} T_{it} Q_{iq} \right]$$

$$+ \sum_{b=1}^{50} \kappa_b B_{ib} + \sum_{t=1}^{30} \phi_t T_{it} + \eta_i$$

- Here B_{ib} is a dummy for birth in state b, T_{it} is a dummy for birth in year t, and Q_{iq} is a dummy for birth in quarter q

Bound, Jaeger and Baker (1995)

- The AK model uses 30 quarter/year and 150 quarter/state interactions as instruments, so is overidentified by 179
- The interactions are meant to capture differences in the effects of compulsory schooling laws across states and over time
- From the point of view of 2SLS asymptotics, adding the interactions can only be a plus: more instruments improve the first stage fit and drive down standard errors
- But the QOB first stage is small and many of the interactions are likely to be irrelevant, so this is a severe many/weak scenario
- Bound, Jaeger and Baker replace the QOB dummies with randomly-generated fake QOB variables that have no first stage by design

Bound, Jaeger and Baker (1995)

	2SLS (real QOB)	2SLS (fake QOB)	OLS
30 QOB*year	0.081	0.062	0.063
interactions	(0.016)	(0.037)	(0.000)
180 QOB*year and state	0.083	0.060	0.063
interactions	(0.009)	(0.015)	(0.000)

Bound, Jaeger and Baker (1995)

- The Bound, Jaeger and Baker results show that the AK 2SLS specification essentially reproduces OLS even if the instruments are junk unrelated to schooling
- Disturbingly, conventional IV standard errors give no indication of the extent of the problem
- The fake QOB 180-instrument model generates an estimate that looks precise despite the lack of a first stage
- How can we diagnose many/weak problems, and correct them if they are present?

Many/Weak Bias

- Consider a model with a single endogenous regressor and Q instruments, written in matrix form:

$$Y = \rho S + \epsilon$$

$$S = Z\pi + \eta$$

- Remember that we can write $\hat{S} = P_Z S$
- We can thus re-write the 2SLS estimator of ρ (regression of Y on $\hat{S}$) as

$$\hat{\rho}_{2SLS} = \rho + (S' P_Z S)^{-1} S' P_Z \epsilon$$

$$= \rho + (S' P_Z S)^{-1} (\pi' Z' + \eta') P_Z \epsilon$$

$$= \rho + (S' P_Z S)^{-1} \pi' Z' \epsilon + (S' P_Z S)^{-1} \eta' P_Z \epsilon$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] = E\left[\left(S'P_Z S\right)^{-1}\pi' Z'\epsilon\right] + E\left[\left(S'P_Z S\right)^{-1}\eta' P_Z \epsilon\right]$$

- It's difficult to make further progress since we cannot pass expectations inside the inverse
- Bekker (1994): passing the expectations through provides an accurate approximation under group asymptotics that allow Q to grow with N
- Then the finite-sample bias of 2SLS can be approximated

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \left(E\left[S'P_Z S\right]\right)^{-1}\pi' E\left[Z'\epsilon\right] + \left(E\left[S'P_Z S\right]\right)^{-1} E\left[\eta' P_Z \epsilon\right]$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \left(E\left[S'P_Z S\right]\right)^{-1} \pi' E\left[Z'\epsilon\right] + \left(E\left[S'P_Z S\right]\right)^{-1} E\left[\eta' P_Z \epsilon\right]$$

- The first term is zero (why)?
- The finite-sample bias of 2SLS comes from the fact that the numerator of the second term is not zero:

$$E\left[\eta' P_Z \epsilon\right] \neq 0$$

- What is the intuition for this?

Many/Weak Bias

- To make further progress, let's use some matrix tricks
  - A scalar equals its trace
  - Trace is invariant to cyclic permutations
  - Trace is a linear operator that passes through expectations

$$E\left[\eta' P_Z \epsilon | Z\right] = E\left[tr\left(\eta' P_Z \epsilon\right) | Z\right]$$

$$= E\left[tr\left(P_Z \epsilon \eta'\right) | Z\right]$$

$$= tr\left(P_Z E\left[\epsilon \eta' | Z\right]\right)$$

$$= \sigma_{\eta\epsilon} tr\left(Z(Z'Z)^{-1}Z'\right)$$

$$= \sigma_{\eta\epsilon} tr\left((Z'Z)^{-1}Z'Z\right)$$

$$= \sigma_{\eta\epsilon} Q$$

Many/Weak Bias

- The denominator of the bias is

$$E\left[S'P_Z S\right] = E\left[(\pi' Z' + \eta')P_Z(Z\pi + \eta)\right]$$

$$= E\left[\pi' Z' Z\pi\right] + E\left[\eta' P_Z \eta\right]$$

- Applying the same trace tricks to the second term shows that $E\left[\eta' P_Z \eta\right] = \sigma_\eta^2 Q$
- Then

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\epsilon} Q}{E\left[\pi' Z' Z\pi\right] + \sigma_\eta^2 Q}$$

$$= \frac{\sigma_{\eta\epsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + E\left[\pi' Z' Z\pi\right] / (\sigma_\eta^2 Q)}$$

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\epsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + E\left[\pi' Z' Z \pi\right] / (\sigma_\eta^2 Q)}$$

- Notice that $E\left[\pi' Z' Z \pi\right]$ is the explained sum of squares from the regression of S on Z, while σ_η^2 is the residual sum of squares divided by N
- Then

$$E\left[\hat{\rho}_{2SLS} - \rho\right] \approx \frac{\sigma_{\eta\epsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1 + F}$$

- Here F is the first stage (population) F-statistic for the joint significance of all the instruments

Many/Weak Bias

$$E\left[\hat{\rho}_{2SLS} - \rho\right] = \frac{\sigma_{\eta\epsilon}}{\sigma_{\eta^2}} \cdot \frac{1}{1+F}$$

- Note also that when F goes to zero, the 2SLS bias becomes $\sigma_{\eta\epsilon}/\sigma_{\eta}^2$
- With a zero first stage, $S = \eta$ so $\sigma_{\eta\epsilon}/\sigma_{\eta}^2 = Cov(S_i, \epsilon_i)/Var(S_i)$, the bias of OLS
- As the first stage shrinks to zero, all of the variance in the fitted values comes from the endogenous part of $\hat{S}$, and the bias of 2SLS equals the bias of OLS

Many/Weak In Practice

- In practice it is a good idea to look at the first stage F-statistic
- In models with exogenous controls this should be the partial F on the excluded instruments
- Stock and Staiger (1997) suggest that many weak typically becomes a problem when $F < 10$, though this is not a theorem
- Standard errors based on Bekker (1994) asymptotics are more accurate in many/weak scenarios (see also Hansen et al. 2008)

Many/Weak In Practice

- The Limited Information Maximum Likelihood (LIML) estimator coincides with 2SLS for just-id models and is approximately median unbiased for overidentified models
- A comparison of LIML and 2SLS can help determine whether many/weak is a problem (though conventional LIML standard errors are too small)
- LIML has some unattractive features, however: it has fat tails (no moments), and is inconsistent with heteroskedasticity
- There are other estimators (e.g. Fuller) that correct some of the deficiencies of LIML; see Stock and Yogo (2005)
- The Anderson-Rubin (AR) test provides a method for testing hypotheses about ρ that is robust to weak instruments, though it has low power when the instruments are weak

The Judges Design

- But when it sensible to talk about a research design with "many" instruments? $\rightarrow$ a research design that uses differences in the harshness/leniency of judges or other examiners is a research design that often has to deal with many/weak potential problems
- Examples: Kling (2006) on criminal sentencing, Doyle (2007) on foster care, Maestas et al. (2013) on disability insurance, Dobbie (2015) on bankruptcy
- If judges are randomly assigned, the identity of the judge is a plausible instrument (there are many fixed effects for judges!)

The Judges Design

- Common approach to the judges design: Define

$$T_i^* = \frac{1}{n_{j(i)} - 1} \sum_{k \neq i} 1\{j(k) = j(i)\} T_k$$

- T_i^* is the mean treatment rate (e.g. incarceration rate) for the judge assigned to i, $j(i)$, for all cases not involving i
- T_i^* is then used as an instrument for i's actual treatment T_i

The Judges Design: JIVE

- Note that T_i^* is the fitted value from a first stage regression of removal on all the judge dummies, leaving out i
- The instruments are implicitly a full set of judge dummies, which may lead to many/weak problems
- Using a leave-out mean changes the estimator from 2SLS to jackknife IV (JIVE), which is one solution to the many/weak problem but not necessarily the optimal one
- Conventional standard errors for JIVE are also incorrect in many/weak scenarios.
- Despite this, many papers use judges design given its implicit attractiveness plus the possibility to summarize nicely heterogenous treatment effects (see the marginal treatment effects literature pioneered by Heckman and Vytacil, 1999,2005).